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Heat stress increases enteric methane emissions yield and intensity while impairing rumen function and productivity in lactating dairy cattle

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ABSTRACT

Emissions by livestock contribute to global warming through greenhouse gases including enteric methane (CH₄). Heat stress (HS) reduces dairy cattle productivity, but its impact on CH₄ emissions is unclear. We hypothesize that HS increases CH₄ intensity in lactating cows. Forty lactating multiparous Holstein cows (mean $\pm$ SD; $\sim 173 \pm 23$ DIM, 703 ± 66 kg of BW, and 46.2 ± 7.6 kg of milk yield) were stratified by lactation number, DIM, and PTA for milk production and then randomly assigned to 2 treatments in a 4-wk crossover design. Twenty cows received treatment sequence of HS abatement via evaporative cooling (shade, fans, and soakers: CL) for 4 wk followed by no cooling (shade only) for another 4 wk, whereas 20 were assigned to a treatment sequence that included 4 wk of no cooling followed by 4 wk of evaporative cooling, all cows had a 2-wk washout period before switching to the opposite treatment. All cows were housed in a freestall barn with Calan gates for daily DMI monitoring and were milked 3 times daily. Enteric gas emissions were measured using a GreenFeed system (GF), and rumination, rumen pH, rumen temperature, and motility were monitored via SmaXtec boluses. The average barn temperature-humidity index during the study was 73.2 (SD = 3.4). Heat stress decreased DMI (19.0 ± 0.3 vs. 24.3 ± 0.3 kg/d). Heat stress decreased rumination time, rumen pH, and rumen motility, whereas ruminal temperature was increased. Consequently, HS reduced milk yield and ECM. Visits of 23/80 replicates were missed due to cows not visiting the GF. Heat stress decreased the daily average number of visits per cow relative to CL cows (2.0 ± 0.1 vs. 1.7 ± 0.1 visits/cow/d). Heat stress reduced daily CH₄ emissions (326.4 ± 8.3 vs. 351.8 ± 7.5 g/d), whereas it increased CH₄ yield (17.0 ± 0.5 vs. 14.5 ± 0.4 , g/kg of DMI) and intensity (11.3 ± 0.5

vs. 9.6 ± 0.4 g/kg of ECM). Present results indicate that HS reduces DMI, which compromises ECM yield, but it also affects rumen function and increases CH₄ yield and intensity, reinforcing its negative effect on the environmental footprint and production efficiency of dairy cattle.

Key words: enteric methane, global warming, rumen function

INTRODUCTION

Heat stress (HS) can compromise the health, welfare, and productivity of homeothermic animals, including cows. This is particularly evident in tropical and subtropical regions, where heat load remains high for most of the year. Dairy cows are especially susceptible because they produce large amounts of metabolic heat, particularly during lactation. Consequently, HS disrupts their thermoregulatory capacity, leading to impaired productivity and overall health (Oliveira et al., 2025).

Feed intake reduction, mediated by the hypothalamic cooling center, is the first response by cows to HS in an attempt to reduce metabolic heat production resulting in lower nutrient availability to sustain greater levels of milk production (Albright and Alliston, 1971; Nascimento et al., 2025). Moreover, mild to severe HS can increase maintenance energy requirements (Wheelock et al., 2010). Hence, increased energy requirements coupled with lower DMI potentially aggravate negative energy balance particularly in early lactation dairy cows resulting in lower milk production (Becker et al., 2019). In addition, mammary epithelial cell turnover and apoptosis increase, while proliferation decreases during HS, further compromising milk production (Tao et al., 2011; Lengi et al., 2022).

Rumen function is also negatively affected by HS. Changes such as decreased rumination time (Müschner-Siemens et al., 2020) and lower pH (Kim et al., 2022) have been reported, potentially due to lower intake and increased lactate bacteria activity as they are more resis-

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The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

tant to higher rumen temperatures compared with other rumen bacteria (Correia Sales et al., 2021). Additionally, salivary flow to the rumen acts as a buffer to maintain optimal ruminal pH; however, during HS exposure, a lower volume of saliva reaches the rumen due to declined rumination time and reduced saliva production to maintain hydration (Kim et al., 2022).

Heat stress-induced reductions in rumen motility may decrease feed passage rate, subsequently increasing retention time. A longer retention time allows more microbial activity on feedstuffs, resulting in increased digestibility; however, if HS is extreme, digestibility declines, as reported by Yadav et al. (2016). This decline might be mediated by the lower pH induced by reduced salivary inflow and increased lactate-producing bacterial activity, which impairs fiber- and starch-digesting bacterial activity (Kim et al., 2022). Additionally, Yadav et al. (2016) also measured CH₄ emissions in heifers exposed to simulated HS where they found lower daily emissions and inconsistent CH₄ yield; however, in this experiment only 4 crossbred heifers were included, and exposure to HS was only for a few hours a day, which potentially explains inconsistencies in results. There are 2 types of methanogens in the rumen, slow- and fast-growing archaea. Most CH₄ is produced by fast-growing archaea, as rumen retention time is usually too low to maintain high populations of slow-growing archaea (Yadav et al., 2013). However, increases in abundance of slow-growing bacteria are expected when under HS due to the increased retention time induced by HS. Indeed, Park et al. (2022) analyzed the microbiota of HS lactating cows compared with a pair-fed thermoneutral group. They report that HS increased abundance of *Methanospaera*, a slow-growing genus; nonetheless, they also observed an unexpected decrease in fast-growing archaea genera, such as *Methanobrevibacter* and *Methanomicrobium*, concluding that more research is needed to understand the dynamics of HS and archaea populations in the rumen.

Despite the physiological, and microbiological effects of HS reported that could be related to CH₄ emissions the direct effects of HS on CH₄ emissions remain relatively unexplored in lactating dairy cattle. To our knowledge, Souza et al. (2023) and Chen et al. (2025) are the only published work to date in dairy cattle; however, they used predictive models to estimate CH₄ emissions from dairy cattle instead of direct measurements. The Souza et al. (2023) model estimated a positive correlation between DMI and daily CH₄ production, and a negative correlation between HS and milk yield, whereas CH₄ intensity increased as HS increased. The Chen et al. (2025) model also estimated an increase in CH₄ intensity as well as lower feed efficiency. However, in vivo studies are needed to confirm the effects of HS on enteric CH₄ emissions. Therefore, we hypothesized that HS increases CH₄

intensity in lactating dairy cattle. The objective of this study was to evaluate the effect of HS on methane emissions measured with a head chamber system installed in the barn and its effects on rumen function using an electronic ruminal bolus. Ultimately, the study aimed to elucidate the relationship between the known effects of HS on DMI and milk yield, and how these responses are linked to rumen function and subsequent methane emission dynamics.

MATERIALS AND METHODS

Cows and Treatments

This experiment was conducted at the University of Florida Dairy Unit with the approval of all procedures by the Institutional Animal Care and Use Committee with approval before beginning the experiment under the protocol # 20220000065. Forty multiparous lactating Holstein cows with an average ($\pm$ SD) of 173 ($\pm$ 23) DIM, 703 ($\pm$ 66) kg of BW, and 46.2 ($\pm$ 7.6) kg of milk yield at the experiment enrollment were randomly assigned to 1 of 2 treatment sequences with 20 cows assigned to 2 treatments per period. Cows were housed in a freestall barn (ceiling height = 5.4 m at lateral sides and 10 m at central highest peak, cows floor space = 9.8 m² not including beds) with sand bedding (recycled sand replaced every 7–10 d) and equipped with automatic individually activated gates for each bunk to measure individual feed intake (Calan Gates System, American Calan, Northwood, NH). The treatment sequences were evaporative cooling followed by HS, or HS followed by evaporative cooling. Cows assigned to HS abatement received evaporative cooling (Figure 1) with a system consisting of fans (with 4 cows/fan, located ~3 m above beds and activated automatically at 20°C. Wind speeds from fans was ~3.6 km/h) and soakers (located above the feed line and activated automatically at 21°C for cycles of 30 s on and 5 min off, and 1 min on and 5 min off after ~24°C; CL), whereas cows assigned to no abatement for HS were housed in an adjacent pen with no CL. Periods lasted 4 wk, and then treatment groups were switched to the opposite treatment for a second 4-wk period resulting in treatment sequences of CL-HS and HS-CL including a 2 wk washout period in between the two 4-wk periods during which all cows were cooled before switching to the opposite treatment. The experiment was a crossover design with twenty 2 $\times$ 2 Latin squares with 2 cows per square. Assignment to squares followed stratification of cows by lactation number, DIM and PTA genetic merit for milk production. Within each block, one cow was assigned to a treatment sequence CL-HS and the other to HS-CL. Temperature and relative humidity in the barns were measured using HOBO U23 Pro data loggers (Onset

Computer Corporation, Bourne, MA) that were placed above the fans and soakers to evaluate environmental conditions in the barns. The study was conducted during the summer of 2024 when temperature average was $27.8^{\circ}\text{C} \pm 1.3^{\circ}\text{C}$ SD (period 1 = 27.9 ± 1.0 , period 2 = 27.6 ± 1.5), relative humidity average was 84.8 ± 6.3 SD (period 1 = 81.7 ± 5.9 , period 2 = 88.0 ± 5.0) and temperature-humidity index (THI) was on average 73.2 ± 3.4 SD (period 1 = 72.4 ± 3.7 , period 2 = 74.6 ± 2.9), a THI that induces HS in cows in the absence of heat abatement. Temperature-humidity index was calculated based on the following equation described by Dikmen et al. (2008): $\text{THI} = (1.8 \times T + 32) - [(0.55 - 0.0055 \times \text{RH}) \times (1.8 \times T - 26)]$, where T = air temperature ($^{\circ}\text{C}$) and RH = relative humidity (%). Cows were fed a diet formulated to meet their requirements according to NASEM (2021; Table 1) ad libitum. Individual feed intakes were calculated subtracting offered feed in the morning and afternoon from residual feed measured after 24 h relative to the morning feeding.

Table 1. Ingredients and chemical composition of TMR and pellets supplied by the GF unit to the cooling and heat stress treated cows during the experiment

Item	TMR	Pellet ¹
Ingredient, % DM		
Corn silage	36.21	
Oat silage	8.62	
Corn meal	18.97	
Soybean hulls ground	6.03	
Soybean meal	10.34	
AminoPlus ²	6.9	
Spectrum Fusion ³	1.21	
Molasses	3.1	
Whole cotton seed	5.17	
Lactating cow mineral mix ⁴	3.1	
Chemical composition, DM basis		
NEL, Mcal/kg	1.73	1.51
CP, %	15.2	11.56
Starch, %	22.04	2.78
NDF, %	36.91	61.98
ADF, %	23.66	45.44
Lignin, %	2.84	1.68
Ether extract, %	4.66	1.56
Ash, %	8.03	5.53

¹Soy hull pellet provided to cows as a bait to visit the GF unit.

²AminoPlus, Ag Proccessing Inc., Omaha, Nebraska.

³Spectrum Fusion, Perdue AgriBusiness, Salisbury, Maryland.

⁴Lactating cow mineral mix contained 12.9% Ca, 2.3% P, 6.0% Mg, 0.1% S, 5.5% K, 11.5% Na, 912 ppm of Zn, 240 ppm of Cu, 387 ppm of Fe, 866 ppm of Mn, 7.6 ppm of Co, 19.2 ppm of I, 9.2 ppm of Se, 134,446 IU of vitamin A/kg, 34,099 IU of vitamin D/kg, and 568 IU of vitamin E/kg (DM basis).

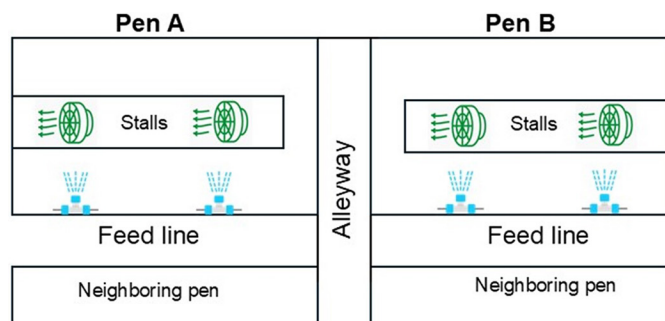


Figure 1. Schematic representation of the evaporative cooling system layout in the experimental pens. All pens were shaded. During period 1, fans (green) and soakers (blue) were off in pen A and on in pen B. During period 2, fans and soakers were off in pen B and on in pen A.

Respiration Rate and Rectal Temperature

Vitals were measured twice per week in all cows at 1400 h to assess the effect of HS, this schedule was chosen to avoid extra stress due to overhandling. Respiration rate was measured by counting thoracic movements for 1 min and rectal temperature was measured using a digital thermometer (Pavia rectal temp, Pavia Sales Group Inc., Minnetonka, MN).

Milk Yield and Milk Components

Cows were milked 3 times per day at 0800, 1900, and 2300 h (unequally spaced due to labor constraints), and milk yield was recorded throughout the study period using AfiLab milk analyzer (Afimilk Agricultural Cooperative Ltd., Kibbutz, Israel). In addition, milk samples were collected twice a week during the first and second milkings (Kamphuis et al., 2016) for a total of 4 samples per week from each cow using 50-mL tubes containing bronopol B-14 pills as a preservative. Milk component analysis included percentages of fat, protein, lactose, and SNF, as well as group of fatty acids percentage (i.e., de novo [$<C$ 16], mixed [16 C] and preformed [>16 C]), MUN (mg/dL), and SCC at the Southeast Milk Inc. Laboratory (Bellevue, FL). Subsequently, ECM was estimated using the following formula: $\text{ECM, kg/d} = [(0.3246 \times \text{milk yield}) + (12.86 \times \text{fat yield}) + (7.04 \times \text{protein yield})]$ and SCC were transformed to SCS as follows $\text{SCS} = [\text{Log}_{10}(\text{SCC}/12.5)]/\text{Log}_{10}(2)$.

Rumen Function Measurements

Rumination time, rumen pH, temperature, and motility data were estimated using the SmaXtec ruminal bolus (SmaXtec Animal Care, Graz, Austria). This device measures 132×35 mm and contains a microprocessor, memory storage, an internal antenna, and a battery. It is

breakproof and resistant to rumen fluid. The bolus contains a pH probe that was calibrated using pH 7 buffer solution before insertion into the animal. Additionally, it is equipped with a triaxial accelerometer that allows the measurement of rumination and rumen motility. The boluses were administered orally into the reticulum using a bolus applicator, following the manufacturer's instructions, with the assumption that they would remain in the reticulum after administration. Rumen pH and temperature were recorded every 10 min, whereas rumination time and rumen motility were measured daily. All data were transmitted wirelessly to a web database for storage and access.

Enteric Gas Measurements

To measure enteric gas output from individual animals, we used 2 GreenFeed (GF) System units (C-Lock Inc., Rapid City, SD). GreenFeed units are automated head chambers that collect short-term measurements of gases from individual cows allowing for onsite measurements from a large number of animals simultaneously without altering the housing or normal management. Animals are attracted to the unit by a small amount of bait feed (i.e., soyhull pellets ~39 g/drop) released from a hopper and bait drops are set as needed across the day. The GF unit integrates airflow, gas concentration, and head position measurements during every visit using an extractor fan and sensors allowing for air sample collections from each animal's visit. Air samples pass through an airflow line equipped with a nondispersive infrared sensor for a continuous gas concentration measurement, which are adjusted for air flow rate, temperature and pressure and corrected for capture rate and background gases. In the present study, the unit was set to allow the cows up to 9 drops per visit, and 6 visits per day with a 3-h interval between visits. The objective of this setting is to obtain high quality visits of around 3 min or more dispersed throughout the day to be considered a representative sample. On average, cows had 1.9 valid

visits per cow per day (SD = 0.6) during the experiment. The gases measured from the animals were CH₄, CO₂, and H₂.

Statistical Analysis

The experiment was a crossover randomized complete block design with repeated measures with individual cow as the experimental unit. Data collected from each cow from d 1 to 28 on each period were averaged into a single mean and used for statistical analysis. Data were analyzed using PROC MIXED of SAS (SAS Institute Inc., Cary, NC). The model to analyze responses with a single mean per cow per period included the fixed effects of treatment (HS vs. CL), period (1 vs. 2), square (1 to 20), and treatment-by-period interaction, and the random effect of cow nested within square. A second model was used to analyze rumen temperature and pH, which were averaged by hour of the day (0000 to 2300 h) for each cow. This model considered fixed effects of treatment, square, hour, period, and treatment by hour interaction, and the random effect of cow within treatment, square, and period. The best model fit was obtained using an autoregressive covariance structure of order 1 (AR(1)). When cows would not visit the GF unit, their values were included as missing values in the gas emissions analysis. Data are presented as LSM with SEM. Significance was considered when $P \leq 0.05$ and tendency when $P \leq 0.10$, and residuals were tested for normality.

RESULTS

Respiration Rate and Rectal Temperature

Treatment with CL reduced ($P < 0.01$) respiration rate (73.3 ± 1.6 vs. 105.2 ± 1.6 breaths/min) and rectal temperature ($P < 0.01$) in cows compared with HS (39.17 ± 0.2 vs. $40.63 \pm 0.2^\circ\text{C}$). In both respiration rate and rectal temperature, a period effect was detected ($P = 0.03$ and $P = 0.01$ respectively; Table 2).

Table 2. Effect of cooling or heat stress treatment (CL or HS) on respiration rate and rectal temperature from cows in mid lactation (CL, n = 40 and HS, n = 40)

Response ²	Treatment				SEM	P-value ¹		
	Period 1		Period 2			Trt	Period	Trt × period
	CL	HS	CL	HS				
Respiration rate, ³ (breaths/min)	69.9	104.5	76.6	105.9	2.3	<0.01	0.03	0.34
Rectal temperature, ⁴ °C	38.86	40.20	39.49	41.06	0.26	<0.01	0.01	0.60

¹Trt = effect of treatment (CL vs. HS); period = effect of period (1 vs. 2); Trt × period = effect of the interaction between Trt and period.

²Vitals were measured twice per week in all cows at 1400 h to assess the effect of HS.

³Respiration rate was measured by counting thoracic movements for 1 min.

⁴Rectal temperature was measured using a digital thermometer (Pavia rectal temp, Pavia Sales Group Inc., Minnetonka, MN).

Table 3. Effect of cooling or heat stress treatment (CL or HS) on DMI, milk yield, and milk components from cows in mid lactation (CL, n = 40 and HS, n = 40)

Response ²	Treatment				SEM	P-value ¹		
	Period 1		Period 2			Trt	Period	Trt × period
	CL	HS	CL	HS				
DMI, kg/d	25.3 ^a	18.9 ^c	23.2 ^b	19.0 ^c	0.4	<0.01	<0.01	0.05
Milk yield, kg/d	38.8 ^a	27.7 ^c	30.8 ^b	25.9 ^c	1.0	<0.01	<0.01	0.03
ECM, kg/d	41.8 ^a	30.0 ^c	34.3 ^b	28.9 ^c	1.1	<0.01	<0.01	0.04
Milk components, %								
Fat	4.07	4.19	4.19	4.24	0.09	0.15	0.16	0.76
True protein	2.94	2.89	3.21	3.05	0.05	<0.01	<0.01	0.44
Lactose	4.1	4.70	4.54	4.49	0.05	0.06	<0.01	0.66
SNF	8.58	8.40	8.64	8.38	0.09	<0.01	0.78	0.71
Fatty acids, %								
<16 C	0.96	0.95	0.99	0.91	0.03	<0.01	0.68	0.31
16 C	1.41	1.54	1.44	1.61	0.04	<0.01	0.05	0.71
>16 C	1.33	1.37	1.37	1.41	0.03	0.07	0.10	0.90
MUN, mg/dL	13.4	13.2	12.3	13.2	0.3	0.07	0.03	0.20
SCS	2.01	2.28	2.68	2.82	0.29	0.30	<0.01	0.85
Milk component yield, kg/d								
Fat	1.57	1.15	1.29	1.10	0.05	<0.01	<0.01	0.06
True protein	1.15	0.81	0.98	0.80	0.03	<0.01	<0.01	0.06
Lactose	1.87 ^a	1.31 ^c	1.40 ^b	1.18 ^c	0.05	<0.01	<0.01	0.02

^{a-c}LSM values in the same row with different superscripts differ ($P < 0.05$) for the Trt × period interaction.

¹Trt = effect of treatment (CL vs. HS); period = effect of period (1 vs. 2); Trt × period = effect of the interaction between Trt and period.

²Milk samples were collected twice a week during the first and second milkings for a total of 4 samples per week from each cow. Samples were analyzed at the Southeast Milk Inc. Laboratory (Bellevue, FL).

DMI and Rumen Function

Heat stress reduced ($P < 0.01$) DMI compared with CL (19.0 ± 0.3 vs. 24.3 ± 0.3 kg/d); however, both the effects of period ($P < 0.01$) and treatment-by-period interaction ($P = 0.05$) were detected (Table 3).

Cooling cows increased ($P < 0.01$) rumination time compared with HS (493.4 ± 9.5 vs. 428.7 ± 9.6 min/d; Figure 2A). Additionally, HS reduced ($P < 0.01$) rumen motility (1.14 ± 0.01 vs. 1.18 ± 0.01 contractions/min; Figure 2C), but increased ($P < 0.01$) ruminal temperature compared with CL (40.6 ± 0.05 vs. $39.8 \pm 0.05^\circ\text{C}$; Figure 2B).

Although no overall treatment effect on ruminal pH was observed when data were analyzed as repeated measures (i.e., hour of the day; $P = 0.31$), effects of time (i.e., hour; $P < 0.01$) and a treatment by time interaction ($P < 0.01$) were observed. The SLICE function of SAS was used to analyze individual hour differences that revealed that HS tended to lower pH at 0500, 0700, 1000, and 1100 h ($P < 0.10$), and reductions at 0800 and 0900 h ($P < 0.05$; Figure 3A). Additionally, ruminal temperature was higher in HS cows throughout the day relative to CL cows, with effects of time ($P < 0.01$) and treatment by time interaction ($P < 0.01$; Figure 3B).

Milk Yield and Milk Components

Providing CL to cows increased ($P < 0.01$) yields of milk (34.8 ± 0.7 vs. 26.8 ± 0.7 kg/d) and ECM ($38.1 \pm$

0.8 vs. 29.5 ± 0.8 kg/d) compared with HS. In both cases, milk and ECM, the effects of period ($P < 0.01$ for both), and treatment-by-period interaction ($P = 0.03$ and $P = 0.04$ respectively) were observed (Table 3). Treatments did not affect milk fat percentage, whereas CL increased ($P < 0.01$) the content of true protein percentage in milk and tended ($P = 0.06$) to increase that of lactose compared with HS. Cows receiving CL had greater ($P < 0.01$) content of SNF but tended ($P = 0.07$) to have less urea N in milk. Providing CL increased ($P < 0.01$) the content of de novo synthesized fatty acids (<16 C) but reduced ($P < 0.01$) that of mixed fatty acids (16 C) and tended to decrease ($P < 0.01$) pre-formed (>16 C). Somatic cell counts were log-transformed to SCS for statistical analysis and no difference was detected between treatments ($P = 0.30$; Table 3). Heat stress reduced the yields of milk fat, milk protein, and milk lactose relative to CL. Period effects were detected ($P < 0.01$) for milk fat, protein, and lactose yields including a treatment-by-period interaction for fat and protein yield and tendency of treatment-by-period interaction for lactose yield (Table 3).

Gas Measurements

Nine out of 40 cows refused to visit during period 1 (CL = 6, HS = 3) and 14 cows refused to visit during period 2 (CL = 1, HS = 13) for a total of 23 out of 80 replicates, whose values were included as missing values

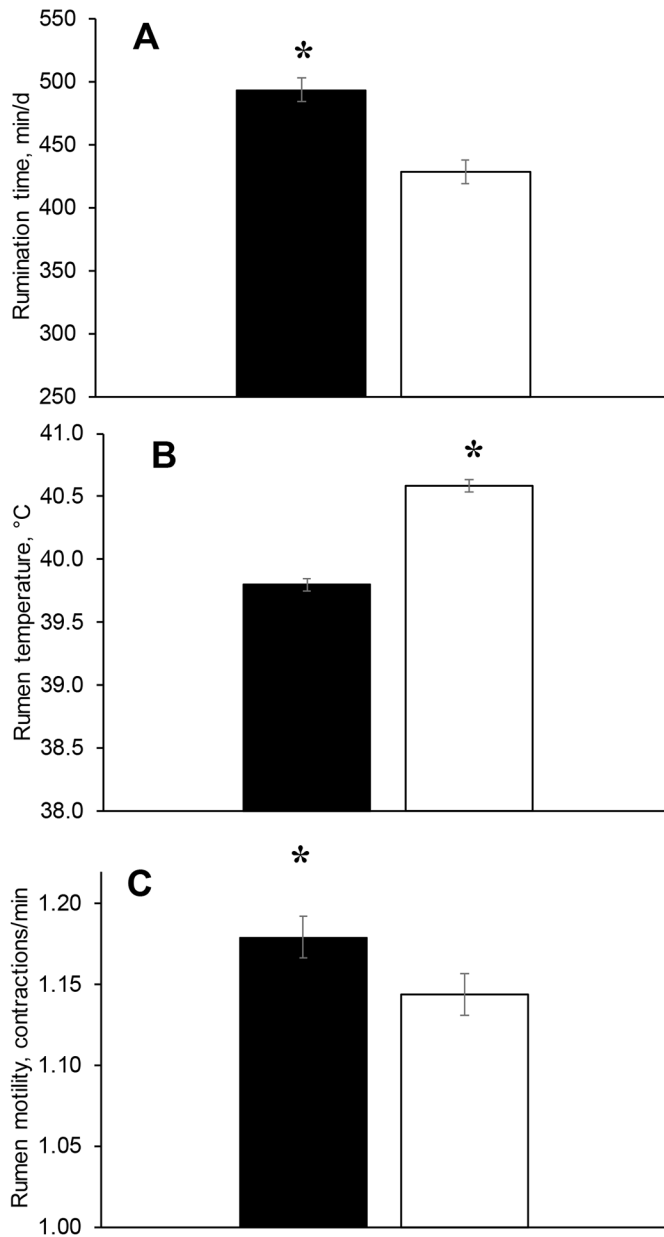


Figure 2. Effects of cooling and heat stress treatment (CL, $n = 40$ or HS, $n = 40$) during two 28-d periods in a crossover arrangement on rumination time ($P < 0.01$; A), rumen temperature ($P < 0.01$; B), and rumen motility ($P < 0.01$; C) from mid-lactation cows. Solid bars represent cows exposed to CL and open bars represent cows under HS. * $P \leq 0.05$.

in the gas emissions analysis. Heat stress decreased ($P < 0.01$) the daily average number of visits per cow relative to CL cows (2.0 ± 0.1 vs. 1.7 ± 0.1 visits/cow per day). A period effect ($P < 0.01$) and treatment-by-period interaction ($P = 0.02$) were also observed (Table 4).

Heat stress lowered ($P < 0.01$) daily CH_4 emissions compared with CL cows (326.4 ± 8.3 vs. 351.8 ± 7.5 g/d). A period effect ($P < 0.01$) and treatment-by-period

interaction ($P < 0.01$) were also observed (Table 4). Conversely, HS increased ($P < 0.01$) methane yield (i.e., g of CH_4 /kg of DMI) relative to CL cows (17.0 ± 0.5 vs. 14.5 ± 0.4 , g/kg of DMI), with a tendency for a period effect ($P = 0.07$) and a treatment-by-period interaction ($P = 0.10$). Heat stress also increased methane intensity (i.e., g of CH_4 /kg of milk or ECM) for both milk yield ($P < 0.01$) and ECM ($P < 0.01$; Table 4). CL induced higher total daily CO_2 emissions ($P < 0.01$) and total daily H_2 emissions ($P < 0.01$) compared with HS cows (Table 4).

DISCUSSION

In this study, we evaluated the effects of HS on lactating dairy cows by comparing 2 groups: one provided with an evaporative cooling system and the other without, during the summer months. We monitored daily DMI, milk yield, rumen function using an electronic ruminal bolus, and methane emissions using a GF unit.

Temperature-humidity index of 64 is considered the borderline between thermoneutral zone and HS for lactating cows producing above 31 kg of milk (Ji et al., 2020). In our study, THI in the barns during the experiment was on average 73.2 ± 3.4 , which is indicative of HS exposure. Cows in the HS treatment responded by increasing rectal temperature as a result of impaired heat dissipation. In addition, HS cows also had increased respiration rate as a compensatory mechanism to increase evaporative heat. Increases in both rectal temperature and respiration rate of HS cows confirm the efficacy of HS abatement via evaporative cooling system.

Removal of heat abatement resulted in reduction of over 5 kg of DMI each day relative to CL cows. Cows responded to the HS insult by decreasing their DMI, as a mechanism to lower heat production from fermentation of feedstuffs (Albright and Alliston, 1971; Nascimento et al., 2025). The difference in DMI between HS and CL cows was 6.4 kg/d in period 1 and 4.2 kg/d in period 2. The greater difference in period 1 relative to period 2 may be attributed to a long-lasting effect of HS, as well as the progression of DIM from period 1 to period 2. Despite the inclusion of a washout period, cows previously exposed to HS during period 1 did not fully recover their DMI when reassigned to the CL group in period 2, which demonstrates the long-lasting negative effects of HS, even when the insult is removed.

As a result of the lower intake of HS cows, rumination time was also decreased by 13% relative to CL cows, which is comparable with previous work from our group in which Toledo et al. (2024) observed an even greater reduction in rumination time due to HS. Lower intake partially explains the reduction in rumination. However, lower rumination time might also be associated with panting as a behavioral response to HS (Grandin, 2016).

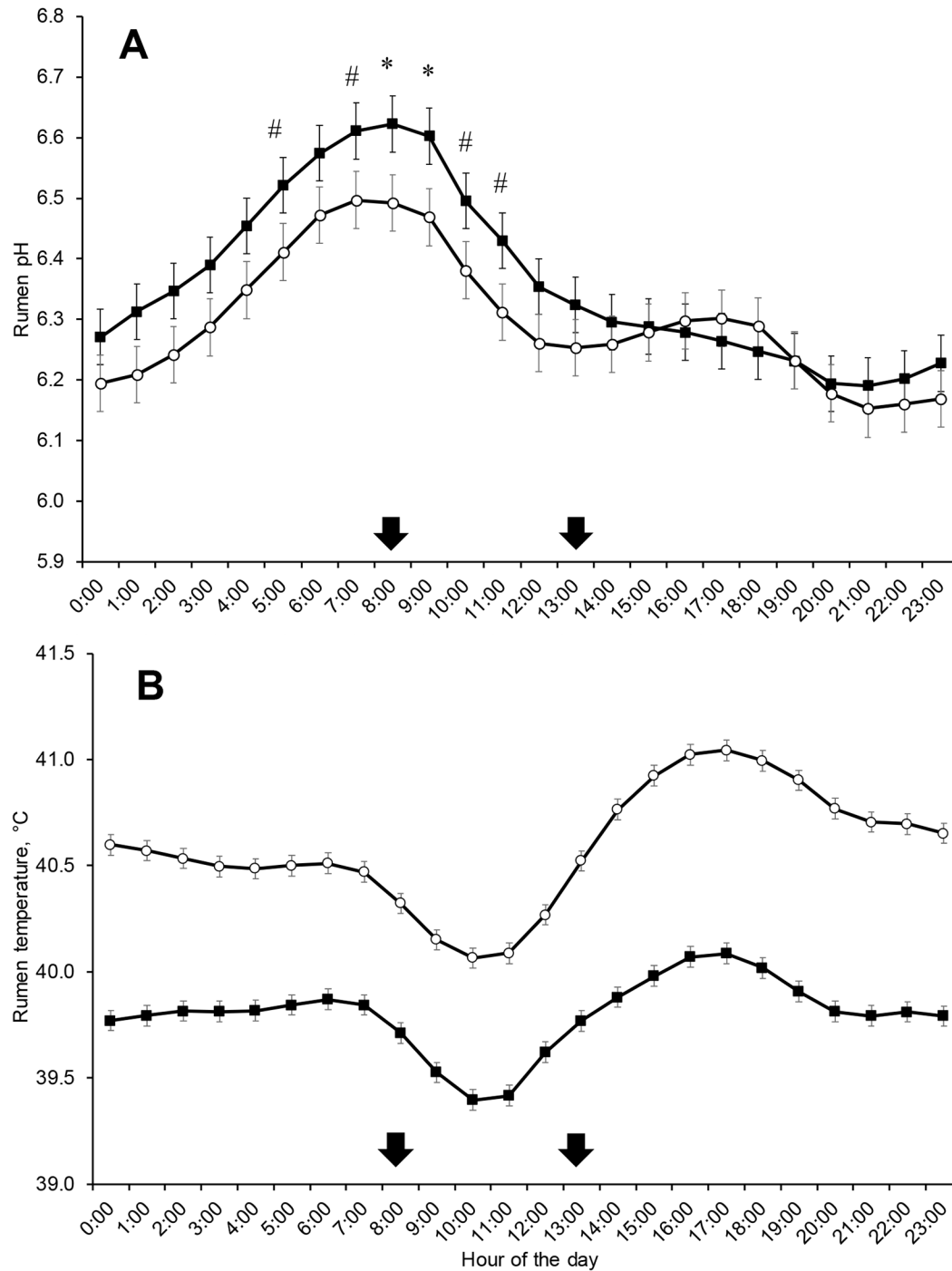


Figure 3. Effects of cooling and heat stress treatment (CL, $n = 40$ or HS, $n = 40$) during two 28-d periods in a crossover arrangement on the circadian pattern of rumen pH ($P = 0.31$; A), and rumen temperature ($P < 0.01$; B) from mid-lactation cows. Solid line with solid squares (■) represents cows exposed to CL, and solid line with open circles (○) represent cows under HS. Arrows represent feeding times. The SLICE function of SAS (SAS Institute Inc., Cary, NC) was used to analyze individual hour differences. * $P \leq 0.05$, # $P \leq 0.10$.

Because of the lower rumination time, ruminal pH was decreased in the HS compared with the CL group, specifically around eating time, because rumination induces salivary buffer to enter the rumen to regulate pH (Kim et

al., 2022). In addition, the combination of panting as a behavioral response to HS and reduced saliva production to maintain hydration during HS further limits salivary buffer reaching the rumen. Moreover, Yadav et al. (2013)

Table 4. Effect of cooling and heat stress treatment (CL or HS) on visits to GreenFeed unit, daily CH₄, CH₄ yield and intensity, and daily CO₂ and H₂ from mid-lactation cows (CL, n = 40 and HS, n = 40)

Response ²	Treatment				SEM	P-value ¹		
	Period 1		Period 2			Trt	Period	Trt × period
	CL	HS	CL	HS				
Visits to GreenFeed unit	2.0 ^a	2.2 ^a	2.0 ^a	1.2 ^b	0.2	<0.01	<0.01	0.02
CH ₄ , g/d	392.2 ^a	319.5 ^b	311.4 ^b	333.3 ^b	14.0	<0.01	<0.01	<0.01
CH ₄ yield, g/kg DMI	15.7	16.9	13.3	17.2	0.8	<0.01	0.07	0.10
CH ₄ intensity, g/kg milk	9.7	11.8	10.3	11.9	0.6	<0.01	0.41	0.71
CH ₄ intensity, g/ kg ECM	9.4	12.0	9.9	10.5	0.8	<0.01	0.34	0.15
CO ₂ , g/d	11,385	10,248	10,975	10,901	321	<0.01	0.50	0.11
H ₂ , g/d	1.65 ^a	0.67 ^b	0.95 ^b	0.92 ^b	0.21	<0.01	0.04	0.04

^{a,b}LSM values in the same row with different superscripts differ ($P < 0.05$) for the Trt × period interaction.

¹Trt = effect of treatment (CL vs. HS); period = effect of period (1 vs. 2); Trt × period = effect of the interaction between Trt and period.

²10 out of the 40 cows refused to visit during period 1 (CL = 7, HT = 3) and 13 cows refused to visit during period 2 (CL = 1, HT = 12) for a total of 23 cows out of 80, whose values were not included in the gas emissions analysis.

and Kim et al. (2022) reported that lactate-producing bacteria have shown improved resistance to high rumen temperatures compared with other rumen bacteria. Following the first feeding at 0800 h, when fresh feed was provided, a sharp decline in ruminal pH was observed in both treatments. In contrast, after the second feeding at 1300 h, the pH did not decrease to the same extent, potentially due to increased heat load during the afternoon, which may have suppressed intake especially in HS cows. During the cooler morning hours, HS cows exhibited a more pronounced reduction in pH, likely attributable to reduced rumination activity and consequently lower salivary buffering capacity. In the afternoon, the mean pH of both treatment groups converged, which may indicate a further reduction in feed intake of HS cows compared with CL cows. Similarly, Toledo et al. (2024) reported reduction in eating time during HS, and Bach et al. (2023) showed that feed intake influences rumen pH.

As with rectal temperature, HS also elevated ruminal temperature. Analysis of the diurnal pattern of ruminal temperature showed effects of treatment, time, and treatment by time interaction. The variation across time is likely attributable to differences in environmental heat load between the lower temperatures at night and morning hours and the hotter temperatures during the afternoon period. Additionally, a decline in ruminal temperature was observed following the first milking at 0800 h, which may be associated with the removal of milk from the mammary gland (Araki et al., 1984; Montes et al., 2023). In contrast, the second (1900 h) and third (2300 h) milkings did not result in a comparable temperature drop, possibly due to the lower milk volumes typically released during these milkings. These findings are consistent with those of Bewley et al. (2008), who reported a strong correlation between rectal and ruminal temperatures using a similar electronic ruminal bolus.

They emphasized the importance of considering milk yield and THI when interpreting temperature data, as high-producing cows generate more metabolic heat than lower-producing counterparts, and increasing THI limits the ability to release heat. Similarly, Ammer et al. (2016) demonstrated that ruminal temperature, measured with a similar ruminal bolus, increased with rising THI, supporting its use as a reliable indicator of HS.

Another important indicator of rumen health is rumen motility, which was reduced by HS relative to CL cows in our study. This finding aligns with previous research by Yadav et al. (2013), who also reported decreased rumen motility under HS conditions. The reduction in motility may be attributed to several factors, including decreased feed intake, and reduced blood flow to the gastrointestinal tract. This redirection of blood flow toward peripheral tissues facilitates heat dissipation but may compromise nutrient delivery and absorption to and from the rumen respectively (Yadav et al., 2013, 2016).

Heat stress accelerates mammary epithelial cell apoptosis (Capuco et al., 2003; Tao et al., 2018), which along with reduced intake and compromised rumen function, resulted in milk yield and ECM reduction by 8 kg/d and 8.6 kg/d, respectively. Lower milk and ECM yields represent losses of 22.9% and 22.6%, respectively, as a consequence of HS. Additionally, in the present study, milk protein percentage decreased under HS treatment, which might also be driving the observed reduction in SNF percentage. Due to the overall decline in milk output, the yields of milk components including fat, protein, and lactose (kg/d), were also reduced in HS cows. Our findings are consistent with those reported by Bernabucci et al. (2010), Tao et al. (2018), and Becker et al. (2019). Heat stress tended to increase MUN, which, together with the reduced milk protein percentage, suggests that cows under HS may experience altered protein

metabolism, potentially due to lower feed intake limiting protein supply, as well as the increased demand of AA to support altered immune function, which further limits the supply to the mammary gland (Loor et al., 2023). Consistently, studies have shown that plasma urea nitrogen also increases under HS, indicating muscle protein breakdown as a compensatory response to decreased protein intake (Bernabucci et al., 2010). Heat stress altered milk fatty acid profiles in this study; de novo fatty acid percentages were reduced, whereas mixed and preformed fatty acid percentages were increased. Our results are consistent with Hammami et al., (2015) and Liu et al. (2017), who also found lower proportions of short-chain and higher proportions of long-chain fatty acids, likely resulting from the observed reduction in feed intake and altered rumen function, which limits the production and absorption of acetate and butyrate, precursors of de novo fatty acids. Similarly to DMI, milk yield, ECM, and milk component yields were affected by both the experimental period and the interaction between treatment and period. As for DMI, the difference in milk yield, ECM, and milk component yields between HS and CL cows were greater in period 1 than in period 2 likely due to lingering effects of HS and advancing DIM despite the washout period inclusion. This difference contributed to the observed period and treatment-by-period interaction highlighting the long-lasting effects and milk losses due to HS. Tao et al. (2018) also reported the long-lasting effects HS on milk production and feed efficiency even further after HS insult was removed.

GreenFeed is a system that measures CH₄ emissions based on voluntary animal visits, which can be a limitation when cows do not visit frequently. In the present study, the GF unit recorded about 2 valid visits per cow per day, even though each cow was allowed to up to 6 visits. A total of 945 valid visits were collected throughout the experiment. In contrast, other studies reported a greater number of visits (Hristov and Melgar 2020). According to manufacturer instructions and evidence from other experiments, variability in visit frequency across studies may be influenced by differences in breed (beef vs. dairy), housing systems, diet, number of animals per GF unit, location, availability of the GF unit, and type of bait used (Alemu et al., 2017; Hristov and Melgar 2020). Additionally, HS reduced the number of visits per cow, especially during period 2, likely due to changes associated with decreased DMI. Both DMI and visits were more affected during period 2, likely because THI was higher during this period. Furthermore, evidence suggests that cows under HS tend to spend less time eating (Toledo et al., 2024), which further supports the lower DMI and reduced visits observed herein. The period and treatment-by-period interaction observed may be explained by the higher THI in period 2 (average visits/cow per day = 1.6;

THI = 74.6) compared with period 1 (average visits/cow per day = 2.1; THI = 72.4). Thus, although the GF system is adaptable to various conditions and can still provide emission estimates, increasing the number of sampling points should remain a priority.

Methane emissions from dairy cattle are influenced by DMI, with a positive correlation indicating that higher feed consumption leads to increased total CH₄ production due to greater substrate availability for rumen fermentation (Bačėninaitė et al., 2022). We observed greater daily CH₄ emissions in CL compared with HS cows, primarily driven by the increased DMI observed in the CL treatment. We also observed effects of period and treatment-by-period interactions, which corresponds with the previously discussed interaction patterns for DMI. Conversely, HS increased CH₄ yield relative to CL cows. Lower rumination time, reduced rumen motility, lower rumen pH, and elevated ruminal temperature induced by HS suggest impaired rumen function, which likely led to longer feed retention time in the rumen. This extended fermentation period may have contributed to increased CH₄ production per unit of DMI, highlighting the link between HS, impaired rumen function and elevated CH₄ yield.

It has been well documented that HS impairs mammary gland function, leading to a reduction in milk yield beyond what would be expected from decreased feed intake alone (Capuco et al., 2003; Tao et al., 2018). Consequently, despite lower intake and reduced milk production, HS cows exhibited higher methane emissions per unit of milk yield and ECM as found in this study.

Unlike methane, CO₂ production is relatively stable and closely linked to energy expenditure, including milk production and physical activity (Alemu et al., 2017). Therefore, given that CL cows had higher feed intake and greater milk yield, the elevated CO₂ emissions observed in this treatment were expected. Because most of the CH₄ in the rumen is produced via the hydrogenotrophic pathway (Hill et al., 2016), higher CH₄ production in CL cows suggests increased H₂ availability, which aligns with the elevated H₂ emissions observed in this group.

Despite efforts to obtain the most accurate results possible, we acknowledge some limitations of the present study. The treatment-by-period interactions observed for DMI, milk yield, and daily CH₄ provide clear evidence of the long-term effects of HS. Although a washout period was included, it may not have been sufficient; therefore, a longer washout period or an alternative study design is encouraged for further HS research. Alternatively, these observations are consistent with a carryover effect of HS on performance even after a return to active cooling, as reported by Tao et al. (2018). As previously discussed, the number of visits to the GF unit, particularly among HS cows, and the number of cows unwilling to visit the

unit could have negatively affected CH₄ emission measurements. Nevertheless, CH₄ emissions were highly correlated with other physiological responses to HS, such as intake and rumen function.

CONCLUSIONS

In the present study, environmental conditions induced HS in the absence of active cooling, as shown by the response of rectal and ruminal temperature and respiration rate of HS-treated cows, confirming the effectiveness of evaporative cooling system. As a response to HS, DMI was reduced, compromising nutrient intake to support milk production. Furthermore, HS decreased rumination time and rumen motility whereas ruminal temperature increased indicating disruption in rumen function. Consequently, milk yield, milk protein percentage, and milk components yield were affected, resulting in substantial production losses extended even after removing HS insult. Finally, HS affected the number of visits to the GF unit. Nonetheless, differences in gas emissions are consistent with intake and ruminal function responses to HS and align with previously reported estimation models. Heat stress increased CH₄ emissions yield and intensity providing further evidence of the negative effects of HS, on productivity and on environmental consequences.

NOTES

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Nonstandard abbreviations used: CL = cooling, GF = GreenFeed; HS = heat stress; THI = temperature-humidity index.

REFERENCES

- Albright, J. L., and C. W. Alliston. 1971. Effects of varying the environment upon the performance of dairy cattle. *J. Anim. Sci.* 32:566–577. <https://doi.org/10.2527/jas1971.323566x>.
- Alemu, A. W., D. Vyas, G. Manafiazar, J. A. Basarab, and K. A. Beauchemin. 2017. Enteric methane emissions from low- and high-residual feed intake beef heifers measured using GreenFeed and respiration chamber techniques. *J. Anim. Sci.* 95:3727–3737. <https://doi.org/10.2527/jas2017.1501>.
- Ammer, S., C. Lambertz, and M. Gauly. 2016. Is reticular temperature a useful indicator of heat stress in dairy cattle? *J. Dairy Sci.* 99:10067–10076. <https://doi.org/10.3168/jds.2016-11282>.
- Araki, C. T., R. M. Nakamura, L. W. G. Kam, and N. Clarke. 1984. Effect of lactation on diurnal temperature patterns of dairy cattle in hot environments. *J. Dairy Sci.* 67:1752–1760. [https://doi.org/10.3168/jds.S0022-0302\(84\)81501-5](https://doi.org/10.3168/jds.S0022-0302(84)81501-5).
- Bačëninaitė, D., K. Džermeikaitė, and R. Antanaitis. 2022. Global warming and dairy cattle; how to control and reduce methane emission. *Animals (Basel)* 12:2687. <https://doi.org/10.3390/ani12192687>.
- Bach, A., M. Baudon, G. Elcoso, J. Viejo, and A. Courillon. 2023. Effects on rumen pH and feed intake of a dietary concentrate challenge in cows fed rations containing pH modulators with different neutralizing capacity. *J. Dairy Sci.* 106:4580–4598. <https://doi.org/10.3168/jds.2022-22734>.
- Becker, C. A., R. J. Collier, and A. E. Stone. 2020. Invited review: Physiological and behavioral effects of heat stress in dairy cows. *J. Dairy Sci.* 103:6751–6770. <https://doi.org/10.3168/jds.2019-17929>.
- Bernabucci, U., N. Lacetera, L. H. Baumgard, R. P. Rhoads, B. Ronchi, and A. Nardone. 2010. Metabolic and hormonal acclimation to heat stress in domesticated ruminants. *Animal* 4:1167–1183. <https://doi.org/10.1017/S175173111000090X>.
- Bewley, J. M., M. E. Einstein, M. W. Grott, and M. M. Schutz. 2008. Comparison of reticular and rectal core body temperatures in lactating dairy cows. *J. Dairy Sci.* 91:4661–4672. <https://doi.org/10.3168/jds.2007-0835>.
- Capuco, A. V., S. E. Ellis, S. A. Hale, E. Long, R. A. Erdman, X. Zhao, and M. J. Paape. 2003. Lactation persistency: Insights from mammary cell proliferation studies. *J. Anim. Sci.* 81(15_suppl_3):18–31. https://doi.org/10.2527/2003.81suppl_318x.
- Chen, L., V. M. Thorup, and S. Ostergaard. 2025. Modeling the effects of heat stress on production and enteric methane emission in high-yielding dairy herds. *J. Dairy Sci.* 108:3956–3964. <https://doi.org/10.3168/jds.2024-25460>.
- Correia Sales, G. F., B. F. Carvalho, R. F. Schwan, L. de Figueiredo Vilela, J. A. Moreno Meneses, M. P. Gionbelli, and C. Luiza da Silva Ávila. 2021. Heat stress influence the microbiota and organic acids concentration in beef cattle rumen. *J. Therm. Biol.* 97:102897. <https://doi.org/10.1016/j.jtherbio.2021.102897>.
- Hammami, H., J. Vandenplas, M. L. Vanrobays, B. Rekik, C. Bastin, and N. Gengler. 2015. Genetic analysis of heat stress effects on yield traits, udder health, and fatty acids of Walloon Holstein cows. *J. Dairy Sci.* 98:4956–4968. <https://doi.org/10.3168/jds.2014-9148>.
- Hill, J., C. McSweeney, A. G. Wright, G. Bishop-Hurley, and K. Kallantar-zadeh. 2016. Measuring methane production from ruminants. *Trends Biotechnol.* 34:26–35. <https://doi.org/10.1016/j.tibtech.2015.10.004>.
- Hristov, A. N., and A. Melgar. 2020. Short communication: Relationship of dry matter intake with enteric methane emission measured with the GreenFeed system in dairy cows receiving a diet without or with 3-nitrooxypropanol. *Animal* 14:s484–s490. <https://doi.org/10.1017/S1751731120001731>.
- Ji, B., T. Banhazi, A. Ghahramani, L. Bowtell, C. Wang, and B. Li. 2020. Modelling of heat stress in a robotic dairy farm. Part 1: Thermal comfort indices as the indicators of production loss. *Biosyst. Eng.* 199:27–42. <https://doi.org/10.1016/j.biosystemseng.2019.11.004>.
- Kamphuis, C., J. K. Burke, S. Taukiri, S. F. Petch, and S. A. Turner. 2016. Shorter sampling periods and accurate estimates of milk volume and components are possible for pasture-based dairy herds milked with automated milking systems. *J. Dairy Res.* 83:326–333. <https://doi.org/10.1017/S0022029916000406>.
- Kim, S. H., S. C. Ramos, R. A. Valencia, Y. I. Cho, and S. S. Lee. 2022. Heat stress: Effects on rumen microbes and host physiology, and strategies to alleviate the negative impacts on lactating dairy cows. *Front. Microbiol.* 13:804562. <https://doi.org/10.3389/fmicb.2022.804562>.
- Lengi, A. J., J. W. Stewart, M. Makris, M. L. Rhoads, and B. A. Corl. 2022. Heat stress increases mammary epithelial cells and reduces viable immune cells in milk of dairy cows. *Animals (Basel)* 12:2810. <https://doi.org/10.3390/ani12202810>.
- Liu, Z., V. Ezernieks, J. Wang, N. W. Arachchilage, J. B. Garner, W. J. Wales, B. G. Cocks, and S. Rochfort. 2017. Heat stress in dairy cattle alters lipid composition of milk. *Sci. Rep.* 7:961. <https://doi.org/10.1038/s41598-017-01120-9>.
- Loor, J. J., V. Lopreato, V. Palombo, and M. D'Andrea. 2023. Physiological impact of amino acids during heat stress in ruminants. *Anim. Front.* 13:69–80. <https://doi.org/10.1093/af/vfad052>.

- Montes, M. E., M. Brunton, A. Mann, K. Teeple, U. George, J. Boerman, and T. Casey. 2023. Relationship between body temperature and behavior of nonpregnant early-lactation dairy cows. *JDS Commun.* 4:308–312. <https://doi.org/10.3168/jdsc.2022-0327>.
- Müschner-Siemens, T., G. Hoffmann, C. Ammon, and T. Amon. 2020. Daily rumination time of lactating dairy cows under heat stress conditions. *J. Therm. Biol.* 88:102484. <https://doi.org/10.1016/j.jtherbio.2019.102484>.
- Nascimento, B. M., K. L. Parker Gaddis, J. E. Koltz, R. J. Tempelman, M. J. VandeHaar, H. M. White, F. Peñagaricano, and K. A. Weigel. 2025. Impact of heat stress on dry matter intake and residual feed intake in mid-lactation dairy cows. *J. Dairy Sci.* 108:7345–7353. <https://doi.org/10.3168/jds.2025-26574>.
- Oliveira, C. P., F. C. de Sousa, A. L. da Silva, E. B. Schultz, R. I. V. Londoño, and P. A. R. de Souza. 2025. Heat stress in dairy cows: Impacts, identification, and mitigation strategies. A review. *Animals (Basel)* 15:249. <https://doi.org/10.3390/ani15020249>.
- Park, T., L. Ma, S. Gao, D. Bu, and Z. Yu. 2022. Heat stress impacts the multi-domain ruminal microbiota and some of the functional features independent of its effect on feed intake in lactating dairy cows. *J. Anim. Sci. Biotechnol.* 13:71. <https://doi.org/10.1186/s40104-022-00717-z>.
- Souza, V. C., L. E. Moraes, L. H. Baumgard, J. E. P. Santos, N. D. Mueller, R. P. Rhoads, and E. Kebreab. 2023. Modeling the effects of heat stress in animal performance and enteric methane emissions in lactating dairy cows. *J. Dairy Sci.* 106:4725–4737. <https://doi.org/10.3168/jds.2022-22658>.
- Tao, S., J. W. Bubolz, B. C. do Amaral, I. M. Thompson, M. J. Hayen, S. E. Johnson, and G. E. Dahl. 2011. Effect of heat stress during the dry period on mammary gland development. *J. Dairy Sci.* 94:5976–5986. <https://doi.org/10.3168/jds.2011-4329>.
- Tao, S., R. M. Orellana, X. Weng, T. N. Marins, G. E. Dahl, and J. K. Bernard. 2018. Symposium review: The influences of heat stress on bovine mammary gland function. *J. Dairy Sci.* 101:5642–5654. <https://doi.org/10.3168/jds.2017-13727>.
- Toledo, I. M., L. T. Casarotto, and G. E. Dahl. 2024. Seasonal effects on multiparous dairy cow behavior in early lactation. *JDS Commun.* 5:379–383. <https://doi.org/10.3168/jdsc.2022-0358>.
- Wheelock, J. B., R. P. Rhoads, M. J. VanBaale, S. R. Sanders, and L. H. Baumgard. 2010. Effects of heat stress on energetic metabolism in lactating Holstein cows. *J. Dairy Sci.* 93:644–655. <https://doi.org/10.3168/jds.2009-2295>.
- Yadav, B., G. Singh, A. K. Verma, N. Dutta, and V. Sejian. 2013. Impact of heat stress on rumen functions. *Vet. World* 6:992–996. <https://doi.org/10.14202/vetworld.2013.992-996>.
- Yadav, B., G. Singh, A. Wankar, N. Dutta, V. B. Chaturvedi, and M. R. Verma. 2016. Effect of simulated heat stress on digestibility, methane emission and metabolic adaptability in crossbred cattle. *Asian-Australas. J. Anim. Sci.* 29:1585–1592. <https://doi.org/10.5713/ajas.15.0693>.